OSHPD Technical Note for Producing Agency for Healthcare Research and Quality Inpatient Mortality Indicators, 2016, 2017, and 2018 Data

Office of Statewide Health Planning and Development

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Table of Contents

Background 1

How are the Inpatient Mortality Indicators useful? 1

Do the Inpatient Mortality Indicators measure actual quality of hospital care? 1

What changes did OSHPD make to the AHRQ Inpatient Quality Indicators software? 2

How do the 2016, 2017, and 2018 Inpatient Mortality Indicator reports compare to the previous reports? 3

What Inpatient Mortality Indicators are included in the 2016, 2017, and 2018 results for California hospitals? 3

How were the AHRQ Inpatient Mortality Indicators calculated? 5

What hospitals are included in the 2016, 2017, and 2018 IMI results? 8

Background

This Technical Note explains how the Agency for Healthcare Research and Quality (AHRQ) Inpatient Quality Indicator (IQI) software was applied to California's patient discharge data collected by the Office of Statewide Health Planning and Development (OSHPD) to generate hospital results for 11 Inpatient Mortality Indicators (IMIs) and five sub-measures for which AHRQ calculates hospital risk-adjusted mortality rates and quality ratings.

The data tables were produced by OSHPD's Healthcare Analytics Branch in the Information Services Division using AHRQ Quality Indicators software Version 2019 for SAS® with the California inpatient data, 2016, 2017, and 2018. OSHPD made California-specific modifications to the software which were discussed with and supported by AHRQ.

The current IMI reports include data from 324 state-licensed general acute care hospitals in 2016, 2017, and 2018. Other AHRQ IQI reports can also be found on the OSHPD website including hospital-level volume and utilization measures.

How are the Inpatient Mortality Indicators useful?

The AHRQ quality indicators and related software, provided at no cost to states, use readily available patient discharge data to highlight possible differences in the quality of care provided by hospitals. These results may provide the foundation for more in-depth analyses of healthcare quality, and are intended to contribute to quality improvement efforts made by hospital administrators, clinicians, quality assurance personnel, and other stakeholders interested in healthcare quality. In addition, when the information is carefully considered along with its limitations and in conjunction with other reliable healthcare provider information, it may inform decisions made by policy makers, patients, and healthcare purchasers.

Do the Inpatient Mortality Indicators measure actual quality of hospital care?

The IMIs are indicators of hospital quality of care but are not definitive determinations of quality. Rather, they are meant to provide a perspective on quality of care inside hospitals by focusing on inpatient mortality for certain procedures and medical conditions. In-hospital mortality rates vary substantially across hospitals, which may provide evidence suggesting that high in-hospital mortality may be associated with deficiencies in the quality of care. Healthcare provdiers can utilize the IMI findings for further investigation and validation of the health outcomes associated with their healthcare practices.

It is also important to note that the hospital IMI resuls for 2016, 2017, and 2018 come with several caveats:

1. California patient discharge data for the reported medical conditions and procedures have not been validated through medical record reabstraction (with a few exceptions) to demonstrate

that patient severity-of-illness and complications are accurately and reliably coded across all hospitals.

2. OSHPD has not performed detailed clinical analyses to identify the processes of patient care that may lead to differential hospital mortality rates.

3. OSHPD has not performed analyses to establish that the risk models for these medical conditions and procedures, using the International Classification of Diseases, Tenth Revision, Clinical Modification/Procedural Classification System (ICD-10-CM/PCS), perform well compared to gold-standard clinical models that include information such as laboratory values, vital signs, and imaging studies.

4. Comparisons of IMIs across years should be made with caution since results prior to 2016 were generated with the ICD-9-CM coding system, while this analysis is based on ICD-10-CM/PCS. Therefore, mortality rates for hospitals may not be comparable across years prior to 2016.

5. Because these indicators use inpatient mortality as the outcome of interest, hospitals that are less likely to transfer patients post-treatment, such as academic medical centers or safety net hospitals, may appear to perform worse. Hospitals that regularly transfer patients to acute care or ventilator-assist facilities may appear to perform better because any deaths occurring post-transfer are not included in their outcome scores.

What changes did OSHPD make to the AHRQ Inpatient Quality Indicators software?

- California uses the statewide observed rate instead of the national reference rate as the benchmark when rating hospitals as “Better” or “Worse”.

- After discussions with AHRQ and University of California researchers, OSHPD staff modified the AHRQ software and implemented confidence intervals (CIs) based on the exact method. All OSHPD outcome reports to date have employed the exact method in calculating CIs. The exact method is based on the exact probability of the number of observed deaths (or a more extreme number) occurring by chance, given the number of expected deaths at a hospital. This approach differs from the normal approximation method used by AHRQ. It relies on fewer distributional assumptions and provides more conservative estimates for hospitals with relatively few expected deaths.[1] AHRQ agreed that the exact method is more appropriate for public reporting and may implement this improvement in future software releases.

- OSHPD reports only unruptured cases for the Abdominal Aortic Aneurysm (AAA) Repair risk-adjusted mortality rate. The unruptured cases include both open and endovascular repair procedures. The exclusion of ruptured cases (open or endovascular) from the risk-adjusted mortality calculation was based on the decision by the former OSHPD Technical Advisory Committee. For patients with ruptured cases, one of the key determinants of survival chance is the time that has elapsed between rupture and the procedure, which is not available in the OSHPD data as a risk adjuster. Therefore, patients with ruptured cases were excluded from the analysis. Further, AHRQ’s version stratified AAA into unruptured open and unruptured

[1] Luft HS, Brown BW Jr. (1993). Calculating the probability of rare events: Why settle for an approximation? Health Services Research, 28, 419-439.

endovascular sub-measures. The number of cases for each sub-measure at most hospitals may be too small to make reporting results reliable. In consultation with AHRQ, a weighted average is used to combine the two sub-measures into one overall unruptured measure after patients' expected mortality rates were calculated using appropriate coefficients. For public reporting purposes, OSHPD utilized the AHRQ performance rating threshold and did not report performance ratings for hospitals with less than three procedures performed on either one of the AAA Repair sub-measures.

How do the 2016, 2017, and 2018 Inpatient Mortality Indicator reports compare to the previous reports?

It is important to note that the reports for 2016, 2017, and 2018 used ICD-10-CM/PCS coding, while reports from previous years used ICD-9-CM. The new ICD-10-CM/PCS codes are very different than ICD-9-CM codes in the fundamental structure and concepts. Thus, comparability with IMI reports prior to 2016 should be made with caution.

In addition, the current reports used Version 2019 of the AHRQ software, while the reports from previous years used Version 5.0 (2015 and 2014), Version 4.5 (2012 and 2013), Version 4.4 (2010 and 2011), Version 4.1a (2008 and 2009) and Version 3.2a (2006 and 2007). Results are not directly comparable to prior years due to different software versions.

Finally, the 2016, 2017, and 2018 reports used the 2016 reference population data and associated regression coefficients, while previous reports used the reference population data and associated regression coefficients from 2012 and before.

What Inpatient Mortality Indicators are included in the 2016, 2017, and 2018 results for California hospitals?

A total of 11 IMIs were calculated and discussed in this report, including five overall measures plus two sub-measures for surgical procedures, and six overall measures plus three sub-measures for medical conditions. As with the previous IMI reports, AAA ruptured cases are excluded from reporting. OSHPD combines unruptured AAA repair cases, open and endovascular, using a weighted average. IMI definitions and technical spectifications can be found on the AHRQ website.

Surgical Procedures:

- **Abdominal Aortic Aneurysm (AAA) Repair, unruptured** – In-hospital deaths per 100 discharges with unruptured AAA repairs, open or endovascular, ages 18 years and older. Excludes ruptured cases, obstetric discharges and transfers to another hospital.
- **Carotid Endarterectomy (CEA)** – In-hospital deaths per 100 discharges with a procedure for carotid endarterectomy, for patients ages 18 years and older. Excludes obstetric discharges and transfers to another hospital.
- **Esophageal Resection** – In-hospital deaths per 100 discharges with a procedure for esophageal resection or total gastrectomy and a diagnosis of esophageal cancer, or with a procedure for esophageal resection and a diagnosis of gastrointestinal cancer, ages 18 years

and older. Excludes obstetric discharges and transfers to another hospital.

- **Pancreatic Resection** – In-hospital deaths per 100 discharges with pancreatic resection, ages 18 years and older. Includes metrics to stratify discharges grouped by presence or absence of pancreatic cancer. Excludes acute pancreatitis discharges, obstetric discharges, and transfers to another hospital.
    - **Pancreatic Resection, Cancer** – Discharges with any diagnosis for total pancreatic resection and pancreatic cancer.
    - **Pancreatic Resection, Other** – Discharges for all other pancreatic resection.
- **Percutaneous Coronary Intervention (PCI)** – In-hospital deaths per 100 discharges with a procedure for percutaneous coronary intervention, for patients 40 years and older. Excludes obstetric discharges and transfers to another hospital.

Medical Conditions:

- **Acute Myocardial Infarction (AMI)** – In-hospital deaths per 100 hospital discharges with AMI as a principal diagnosis for patients ages 18 years and older. Excludes cases in hospice care at adminission, obstetric discharges, and transfers to another hospital.
- **Acute Stroke** – In-hospital deaths per 100 hospital discharges with acute stroke as a principal diagnosis for patients ages 18 years and older. Includes metrics for discharges grouped by type of stroke. Excludes cases in hospice care at admission, obstetric discharges, and transfers to another hospital.
    - **Acute Stroke, Hemorrhagic** – Discharges with a principal diagnosis code for intracerebral hemorrhagic stroke.
    - **Acute Stroke, Ischemic** – Discharges with a principal diagnosis code for ischemic stroke.
    - **Acute Stroke, Subarachnoid** – Discharges with a principal diagnosis code for subarachnoid hemorrhage.
- **Gastrointestinal (GI) Hemorrhage** – In-hospital deaths per 100 discharges with a principal diagnosis of gastrointestinal hemorrhage; or a secondary diagnosis of esophageal varices with bleeding along with a qualifying associated principal diagnosis, for patients ages 18 years and older. Excludes obstetric discharges, cases in hospice care at adminission, discharges with a procedure for liver transplant, and transfers to another hospital.
- **Heart Failure** – In-hospital deaths per 100 hospital discharges with heart failure as a principal diagnosis for patients ages 18 years and older. Excludes cases in hospice care at admission, obstetric discharges and transfers, to another hospital.
- **Hip Fracture** – In-hospital deaths per 100 hospital discharges with hip fracture as a principal diagnosis for patients ages 65 years and older. Excludes periprosthetic fracture discharges, obstetric discharges, cases in hospice care at admission, and transfers to another hospital.
- **Pneumonia** – In-hospital deaths per 100 hospital discharges with pneumonia as a principal diagnosis for patients ages 18 years and older. Excludes obstetric discharges, cases in

hospice care at admission, and transfers to another hospital.

Two measures, included in the 2019 version of the AHRQ software, were excluded from OSHPD reporting.

- Coronary artery bypass graft (CABG) surgery – the OSHPD California CABG Outcomes Reporting Program (CCORP) already reports hospital and surgeon-level risk-adjusted mortality rates and quality ratings using data from a clinical registry expressly created for quality monitoring and reporting. This, along with other features of the data collected by CCORP, results in superior quality assessments to those obtained from the AHRQ CABG IMI measure.

- Acute myocardial infarction (AMI) – AHRQ IMIs include two measures: one includes all AMI patients and one excludes patients transferred to another acute care hospital. Upon advice from experts on its former TAC, OSHPD decided to report only the measure that includes transfer patients. Analyses show that transfer patients were, on average, less severely ill and experienced lower mortality rates than non-transfer patients, so hospitals that received large numbers of transfer patients were not disadvantaged by this decision.

How were the AHRQ Inpatient Mortality Indicators calculated?

OSHPD used a modified version of the AHRQ Quality Indicators software Version 2019 for SAS, released in August 2019. AHRQ's free software and associated documentation are available online at http://www.qualityindicators.ahrq.gov/software/SAS.aspx.

The first step in calculating rates was to transform the data elements and values of the California patient discharge data into a format that can be read by the AHRQ software. Second, OSHPD specified the number of diagnoses and procedures available in the data set. Third, All Patient Refined Diagnosis Related Groups (APR-DRG) "groupers" and associated "risk of mortality" categories were added to each patient record by running the 3M Health Information Systems Corporation software licensed to AHRQ. Finally, the coefficients used in the risk-adjustment process (described below), as well as population rates, were constructed based on the 2016 National Inpatient Sample (NIS) compiled by the AHRQ Healthcare Cost and Utilization Project (HCUP). The coefficients from the 2016 NIS were used for the IMI report for 2016-2018. Once the data were transformed and the options were set, the software was run to automatically calculate the rates described below.

Standardizing the Patient Data

California hospitals electronically submit inpatient data including patient age, length of stay, gender, race, ICD-10-CM/PCS codes, and related information to OSHPD. The OSHPD Healthcare Information Division Patient Data Section then applies thousands of quality control automated "edits" using a custom software program that flags data values submitted by hospitals to OSHPD as invalid or likely wrong. If certain thresholds are reached, hospitals are contacted and asked to review the data and make any necessary changes. Once the data have been finalized, OSHPD researchers use SAS software to transform the data elements to conform to the standards specified in the AHRQ documentation. These are the same standards that AHRQ applies to the State Inpatient Database and the NIS, collected from most states and maintained by the federal government.

Calculation of Observed Rates

The AHRQ IMI software produces numerators, denominators, observed rates, expected rates, risk-adjusted rates, and additional information to evaluate confidence intervals and reliability of the indicators. The current report produced by OSHPD focused on risk-adjusted rates and confidence intervals for California acute care hospitals. Terminology and methodology used for determining these rates are described below to help explain the process of generating risk-adjusted rates.

Numerator – The number of inpatient deaths that occurred in a specific denominator population. For example, the number of patients who died within the hospital after being admitted for heart failure (after excluding patient records based on the denominator definition).

Denominator – For each IMI, expert clinicians used ICD-10-CM/PCS codes to select patient discharge records with diagnoses or procedures that indicate a particular condition or procedure. For example, heart failure is a complex condition that can be defined by numerous diagnoses, thus clinicians select only the specific codes that represent the intended concept of the indicator. From the initial cohorts of patients, some records were excluded. For example, patients that were transferred to another short-term hospital were excluded for some cohorts (see AHRQ documentation for additional exclusion criteria). In addition, maternal patients were excluded when constructing most of the indicators. In sum, the denominators represent the total number of patients for specific conditions or procedures that are "at risk" of dying during their hospital stay.

Observed Rates – An observed mortality rate is defined as the number of patient deaths that occur within a specified group of patients admitted to the hospital for a medical condition or surgical procedure.

Calculation of Expected Deaths at Each Hospital

The purpose of statistical risk adjustment is to provide an equitable comparison between hospitals by accounting for hospitals that treat sicker patients versus those that treat healthier ones. To make comparisons fair, it is necessary to hold the patient "case mix" of hospitals constant by adjusting for the illness severity of patients. To create risk-adjusted rates, the first step is to estimate how many people would be expected to die in a particular hospital if they had a mix of patients that was comparable to the average hospital from the reference population (the 2016 California observed rates for this report). Although the particular methods require technical expertise, the process of generating expected rates is straightforward.

Step 1: Select Risk Factors to Predict Inpatient Death

Consulting with medical experts and statisticians, AHRQ chose risk factors that predicted hospital inpatient death. For most of the IMIs, the risk factors include patient age, gender, procedure/condition category, and a risk-of-mortality score associated with each procedure/condition category. To assign each patient into a procedure/condition category, AHRQ selected a proprietary tool from the 3M Health Information Systems Corporation—the All Patient Refined Diagnosis Related Groups (APR-DRGs). The APR-DRG system works with hospital administrative data and provides a way to categorize patients into procedure/condition groups, and, given membership to that group, to estimate the severity of patients' diseases and the likelihood that they will die in the hospital. These estimates are calculated by looking at patient age, principal diagnosis, and secondary diagnoses to

assign each patient into one of four categories (low, moderate, high, or very high) for disease severity and risk of mortality.

OSHPD staff used the AHRQ-licensed software from 3M to apply the APR-DRG fields to the standardized California hospital inpatient data described above. This creates the base APR-DRG category and the associated "risk of mortality" fields in the dataset.

Step 2: Create Multivariate Model to Predict Inpatient Death

Logistic regression models were built by AHRQ to predict patient's probability of death. When possible, hospital-level models are estimated using General Estimating Equations (GEE) (hierarchical modelling) to account for within-hospital correlation. If the GEE model does not converge, a logistic regression model is used. Each indicator has a set of covariates identified in a l]ogistic regression model. Its risk adjustment parameters are estimated based on population data. AHRQ has published a more detailed summary of how these models work on its website (http://www.qualityindicators.ahrq.gov/modules/Default.aspx).

Step 3: Apply Model Coefficients to California Data to Calculate Predicted Probability of Death

The software provided by AHRQ includes the coefficients for each IMI that were created by the multivariate model on the 2016 NIS. To enable custom reports on new samples of data, the AHRQ software identifies which risk factor is present for each patient. Then the coefficients are appropriately applied to the California data so that a predicted probability of death is assigned to each patient. The predicted probability calculated from this step is also referred to as the "direct predicted rate."

Step 4: Estimate Expected Deaths at Each Hospital

The first three steps assign a probability of death for each patient record. To obtain the expected number of deaths for each hospital, the software simply adds up all of the patient-level probabilities for each facility.

Calculation of Risk-Adjusted Rates

With observed and expected mortality rates available for each hospital, it is then possible to construct risk-adjusted rates. While it is sufficient to compare the difference between observed and expected rates to assess higher and lower quality, adding a reference population makes it easier to compare rates. The risk-adjusted (or indirectly standardized) death rate at a hospital equals the State Observed Rate, multiplied by the ratio of the number of observed deaths to the number of expected deaths at that hospital (Observed Cases/Expected Case or "O/E" ratio). The O/E ratio provides a transparent and easy-to-understand assessment of that hospital's performance. A ratio that is less than one indicates there were fewer actual deaths than expected (a good result) while a ratio greater than one indicates that there were more deaths than would be expected given the level of risk in the patient mix.

Calculation of Statistical Outliers

For each IMI, hospitals were rated as "better than expected" if their risk-adjusted death rates were significantly lower than the statewide observed rate. They were rated as "worse than expected" if their rates were significantly higher than the statewide risk-adjusted rate of the particular IMI. To

calculate outlier ratings, OSHPD used the 95% upper and lower confidence intervals. The Version 2019 of the AHRQ software calculates confidence intervals (CI) using the normal approximation as follows:

Lower CI = “Hospital A” risk-adjusted rate – (1.96 * Standard Error)
Upper CI = “Hospital A” risk-adjusted rate + (1.96 * Standard Error)

The standard error for the risk-adjusted rate (for each hospital) is based on the following formula:

The Root Mean Squared Error (RMSE) for each hospital is:

RMSE = square root (“Hospital A” risk-adjusted rate * (1 – “Hospital A” risk-adjusted rate))

The Standard Error is:

SE = RMSE / square root (“Hospital A” denominator)

For example:

If “Hospital A” had a rate of 0.20 and a denominator of 500:

Lower CI = 0.20 – 1.96 * sqrt [(0.20 * (1 – 0.20)) / 500]
Upper CI = 0.20 + 1.96 * sqrt [(0.20 * (1 – 0.20)) / 500]

OSHPD employed the exact method in calculating confidence intervals (CIs) to provide more conservative estimates for hospitals with relatively few expected deaths.

To identify statistical outliers, OSHPD compared hospital risk-adjusted rates to the upper and lower CIs. If a hospital’s upper CI is less than the statewide observed rate, it is designated as performing “better” than the average hospital. If a hospital’s lower CI is greater than the state rate, it is designated as performing “worse” than the average hospital. Using this approach, one can be 95% confident that a rating of “better than expected” or “worse than expected” was not obtained by chance. Smaller hospitals, however, have less statistical power to be classified as performance outliers, especially significantly “better” than the statewide rate. Their risk-adjusted death rates would have to be much higher or lower than a high-volume hospital’s for them to be significantly different from the state average. Conversely, a large hospital with more patients for a particular indicator may be identified as significantly different even when its death rate differs only moderately from the state average.

What hospitals are included in the 2016, 2017, and 2018 IMI results?

Data used for this analysis came from California state-liscensed general acute care hospitals. Certain hospitals were excluded from reporting due to the exclusion crtieria as follows. Based on the exclusion criteria, a total of 324 hospitals were reported in the OSHPD 2016-2018 IMI results. A list of hospitals excluded from the IMI results and the exclusion criteria are presented in Table 1.

- Hospitals identified by the Centers for Medicare and Medicaid Services (CMS) as long-term

acute care hospitals, or hospitals having an average length of stay that exceeded CMS-designated long-term acute care hospitals were excluded from this analysis. These hospitals treat patients with long-term acute conditions (e.g., requiring respiratory care) and have an average length of stay greater than 25 days.

- Hospitals specializing in pediatric care were excluded from this report.

- Hospitals with fewer than three patients reported for each of the 11 conditions and procedures were excluded from this report. The AHRQ software does not report results for a specific IMI if there were two or fewer cases in the denominator for a given hospital. Therefore, hospitals with fewer than three cases in the denominator for all indicators are not included in the report.

Note that if a hospital had a name change between years, the discharges were attributed to the name of the hospital in use at the time the services were provided. Tables 2, 3, and 4 show hospitals that changed names between 2015 and 2016, between 2016 and 2017, and between 2017 and 2019, respectively.

Table 1. Hospitals excluded from 2016, 2017, and 2018 AHRQ IMI results due to provision of long-term acute care (CMS determination), pediatric facility designation, or fewer than three patients reported for all AHRQ IMIs

Type of Exclusion	Hospital Name	Years Excluded From
CMS Long-term Acute Care	Barlow Respiratory Hospital	2016, 2017, and 2018
	Central Valley Specialty Hospital	2016, 2017, and 2018
	Foothill Regional Medical Center	2016, 2017, and 2018
	Kentfield Hospital	2016, 2017, and 2018
	Kindred Hospital – Baldwin Park	2016, 2017, and 2018
	Kindred Hospital – Brea	2016, 2017, and 2018
	Kindred Hospital – La Mirada	2016, 2017, and 2018
	Kindred Hospital – Los Angeles	2016, 2017, and 2018
	Kindred Hospital – Ontario	2016, 2017, and 2018
	Kindred Hospital – Rancho	2016, 2017, and 2018
	Kindred Hospital – Riverside	2016, 2017, and 2018
	Kindred Hospital – San Diego	2016, 2017, and 2018
	Kindred Hospital – San Francisco Bay Area	2016, 2017, and 2018
	Kindred Hospital – South Bay	2016, 2017, and 2018
	Kindred Hospital – Westminster	2016, 2017, and 2018
	Los Angeles County/Rancho Los Amigos National Rehabilitation Center	2016, 2017, and 2018
	Monrovia Medical Center	2016, 2017, and 2018
	Promise Hospital of East Los Angeles – East Los Angeles Campus	2016 and 2017
	Promise Hospital of San Diego	2016 and 2017
	San Joaquin Valley Rehabilitation Hospital	2016, 2017, and 2018
	Vibra Hospital of Northern California	2016, 2017, and 2018
	Vibra Hospital of Sacramento	2016, 2017, and 2018
	Vibra Hospital of San Diego	2016, 2017, and 2018
Pediatric Facility	Children's Hospital and Research Center at Oakland	2016, 2017, and 2018
	Children's Hospital at Mission	2016, 2017, and 2018
	Children's Hospital of Los Angeles	2016, 2017, and 2018
	Children's Hospital of Orange County	2016, 2017, and 2018
	Earl and Loraine Miller Children's Hospital	2016 and 2017
	Loma Linda University Children's Hospital	2016, 2017, and 2018
	Lucile Salter Packard Children's Hospital at Stanford	2016, 2017, and 2018
	Martin Luther King Jr. – Harbor Hospital	2016, 2017, and 2018
	Memorialcare Miller Children's & Women's Hospital Long Beach	2018
	Rady Children's Hospital – San Diego	2016, 2017, and 2018
	Sharp Mary Birch Hospital for Women and Newborns	2016, 2017, and 2018
	Shriners Hospital for Children	2016 and 2017
	Shriners Hospital for Children Northern California	2016, 2017, and 2018
	Sutter Maternity and Surgery Center of Santa Cruz	2016, 2017, and 2018
	University of Southern California Kenneth Norris, Jr. Cancer Hospital	2016, 2017, and 2018
	Valley Children's Hospital	2016, 2017, and 2018
	Children's Recovery Center of Northern California	2018

Fewer Than Three Patients	Catalina Island Medical Center	2016, 2017, and 2018
	College Hospital Costa Mesa	2016, 2017, and 2018
	Colusa Regional Medical Center	2017
	Fresno Surgical Hospital	2016, 2017, and 2018
	Laguna Honda Hospital and Rehabilitation Center	2018
	Menlo Park Surgical Hospital	2016, 2017, and 2018
	Miracle Mile Medical Center	2016, 2017, and 2018
	Motion Picture and Television Hospital	2016
	Patients' Hospital of Redding	2016, 2017, and 2018
	Southern Inyo Hospital	2016 and 2017
	Stanislaus Surgical Hospital	2016, 2017, and 2018
	Surprise Valley Community Hospital	2016, 2017, and 2018
	Sutter Surgical Hospital – North Valley	2016, 2017, and 2018
	Telecare Stanislaus County Psychiatric Health Facility	2017 and 2018

Table 2. Hospitals with Name Changes between 2015 and 2016

Hospital Name in 2015	Hospital Name in 2016
Tehachapi Hospital	Adventist Health Medican Center Tehachapi Valley
Casa Colina Hospital and Centers for Rehabilitative Medicine	Casa Colina Hospital
San Francisco General Hospital	Prescilla Chan & Mark Zuckerberg San Francisco General Hospital & Trauma Center
Riverside County Regional Medical Center	Riverside University Health System – Medical Center
San Antonio Community Hospital	San Antonio Regional Hospital
Stanford Hospital	Stanford Health Care
Doctors Hospital of West Covina, Inc.	West Covina Medical Center

Table 3. Hospitals with Name Changes between 2016 and 2017

Hospital Name in 2016	Hospital Name in 2017
Feather River Hospital	Adventist Health Feather River
Adventist Medical Center – Reedley	Adventist Health Reedley
San Joaquin Community Hospital	Adventist Health Bakersfield
Adventist Health Medical Center Tehachapi Valley	Adventist Health Tehachapi Valley
Adventist Medical Center	Adventist Health Hanford
Saint Helena Hospital – Clearlake	Adventist Health Clearlake
Glendale Adventist Medical Center	Adventist Health Glendale
White Memorial Medical Center	Adventist Health White Memorial
Frank R. Howard Memorial Hospital	Adventist Health Howard Memorial
Ukiah Valley Medical Center	Adventist Health Ukiah Valley
Saint Helena Hospital	Adventist Health Saint Helena
Kaiser Foundation Hospital – San Diego	Kaiser Foundation Hospital – San Diego – Zion
Lodi Memorial Hospital	Adventist Health Lodi Memorial
Kaiser Foundation Hospital – Rehabilitation Center Vallejo	Kaiser Foundation Hospital & Rehabilitation Center – Vallejo
Memorial Hospital Medical Center – Modesto	Memorial Medical Center – Modesto
Sonora Regional Medical Center – Greenley	Adventist Health Sonora – Greenley
Simi Valley Hospital and Healthcare Services – Sycamore	Adventist Health Simi Valley

Table 4. Hospitals with Name Changes between 2017 and 2018

Hospital Name in 2017	Hospital Name in 2018
Citrus Valley Midical Center – Inter Community Campus	Emanate Health Inter – Community Hospital
Citrus Valley Midical Center – Queen of the Valley Campus	Emanate Health Queen of the Valley Hospital
Colusa Regional Medical Center	Colusa Medical Center
Foothill Presbyterian Hospital – Johnston Memorial	Emanate Health Foothill Presbyterian Hospital
Long Beach Memorial Medical Center	Memorialcare Long Beach Medical Center
Marina Del Rey Hospital	Cedars-Sinai Marina Del Rey Hospital
Orange Coast Memorial Medical Center	Memorialcare Orange Coast Medical Center
Palm Drive Hospital	Sonoma Specialty Hospital
PIH Hospital – Downey	PIH Health Hospital – Downey
Presbyterian Intercommunity Hospital	PIH Health Hospital – Whittier
Rideout Memorial Hospital	Adventist Health and Rideout
Saddleback Memorial Medical Center	Memorialcare Saddleback Medical Center
Valleycare Medical Center	Stanford Health Care – Valleycare